

AI-Powered Artwork Analysis Tasks

The candidate must build two independent AI-based pipelines that analyse artwork images and return structured outputs. Both tasks must be submitted as a single working demo (API, script, or notebook).

Reference artwork

Lot 242 — Suduca Auction House, Toulouse
<https://www.suduca.com/vente/tableaux-estampes-lot-242-2/>
Contains two lithographs: Guy DUC — "La Pythonisse" (artist's proof, pencil signed lower right, plate-titled) and Marie José LLORCA — "Grange brûlée à Dammartin sous Tigeaux" (pencil signed lower left, numbered 147/280, plate-dated 1972). Task 1 uses only the first artwork image. Task 2 uses all available images.

Artwork content extraction

- 1
- Analyse the primary artwork image and generate structured descriptive metadata from its visual content.

DESCRIPTION

Given a single artwork image as input, the system must use a vision-capable AI model to understand the visual content — subject matter, style, mood, composition, colours, and technique — and return three distinct pieces of text metadata about it.

DELIVERABLES

Keywords	One-liner caption	Description
5–10 concise tags capturing subject, style, medium, mood, and era. Suitable for search indexing.	A single sentence (≤20 words) describing what is depicted — readable by a general audience.	3–5 sentences covering composition, technique, subject, mood, and notable details. Suitable for an auction catalogue.

CONSTRAINTS

- Input must be an image URL or base64-encoded image — not manually described text.
- All three outputs must be derived from the visual content of the image only; do not rely on external metadata, filenames, or URLs.
- Output must be structured (JSON recommended) with clearly labelled fields: *keywords*, *caption*, *description*.

■ Focus only on the first/primary artwork image. Ignore watermarks, auction house logos, or framing artifacts.

Focus image: Use only Guy DUC's "La Pythonisse" lithograph (the first image of the lot) for Task 1. The output should reflect the visual content of that specific artwork alone.

Artist signature extraction

2 Identify and isolate only the artist's handwritten or hand-drawn signature from one or more artwork images, ignoring all other text present.

DESCRIPTION

Artwork images often contain multiple text elements: plate titles, edition numbers, date inscriptions, catalogue notes, stamps, publisher names, and the artist's actual signature. The system must process all provided images and return only the text that constitutes the artist's signature — nothing else.

DELIVERABLES

Signature text	Location hint	Confidence score
The extracted text of the artist's signature exactly as it appears — preserving capitalisation and abbreviations.	Where the signature was found, e.g. "lower right, pencil, below image margin".	A value 0–1 representing how confident the system is that the extracted text is genuinely an artist signature.

KEY CHALLENGE — WHAT MUST BE EXCLUDED

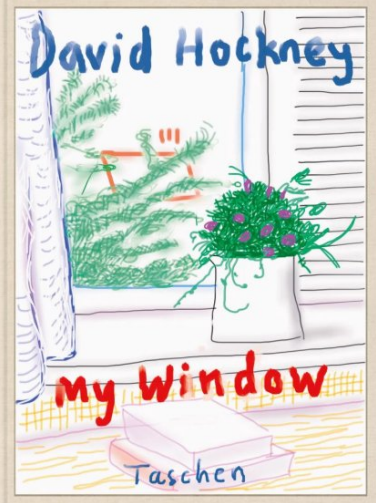
- **Plate/block text:** Titles printed within the lithographic plate (e.g., "La Pythonisse") must be excluded — these are part of the artwork composition, not the signature.
- **Edition / justification numbers:** Numbering such as "147/280" indicating print run must not be returned as a signature.
- **Date inscriptions:** Years or dates written in the plate or margin (e.g., "1972") must be excluded.
- **Annotations and markings:** Labels like "E.A." (épreuve d'artiste), stamps, or house watermarks must not be mistaken for a signature.
- **Titles written in pencil:** When an artwork is both titled and signed in pencil below the image, only the signature portion must be returned — not the title.
- **Publisher / brand names:** Printed names like publisher imprints or gallery stamps must be excluded even when written in a stylised handwritten font.

REFERENCE IMAGES

Reference
images for
Task 2

- 1. Suduca Lot 242 (both lithographs):
<https://www.suduca.com/vente/tableaux-estampes-lot-242-2/>
- 2. Bonhams — Marco Antonio Gomez, "Winter in the Land":
<https://www.bonhams.com/auction/21781/lot/3030/marco-antonio-gomez-american-1910-1972-winter-in-the-land-30-x-20in/>
Signed lower right in paint. Contains no other text elements — provides a clean single-signature baseline test case.

EXAMPLE IMAGE — SIGNATURE VS. OTHER TEXT

The image is a drawing by David Hockney. At the top, the name 'David Hockney' is written in blue, handwritten-style capital letters. In the center, there is a window with a white frame. Inside the window, a green plant with red flowers is visible. Below the window, there is a white vase with a green plant. At the bottom of the drawing, the words 'my window' are written in red, lowercase letters. Below that, the word 'Taschen' is written in blue, lowercase letters. The background of the drawing is yellow with some green and red lines.

Example image — David Hockney, "My Window" (Taschen)

This image contains three distinct text elements: **"David Hockney"** (top, in blue handwritten-style lettering — this IS the artist signature/name to extract), **"my Window"** (centre-bottom, in red — this is the artwork/book title, NOT the signature), and **"Taschen"** (bottom, in blue — the publisher's name, NOT the signature).

The system must return only *"David Hockney"* and correctly discard the title and publisher text.

All images: Task 2 must process all available images of the lot (both artworks). Each image may yield a separate signature result. Return a list — one entry per detected signature.

Evaluation criteria

Accuracy Does Task 1 correctly describe the artwork? Does Task 2 return only the signature and nothing else?	Noise rejection How well does Task 2 filter out non-signature text (titles, numbers, dates, publisher names, stamps)?
Output structure Is the output clean, consistent JSON? Are all required fields present and correctly typed?	Prompt engineering Is the AI prompt well-designed, specific, and reproducible? Is it robust to edge cases?

Code quality Is the implementation clean, modular, and easy to extend to other artworks or image sets?	Justification Can the candidate explain their design choices — model selection, prompt structure, filtering logic?
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Expected output format: JSON response • Image URL or base64 as input • Structured fields per task

- List output for Task 2 (one entry per detected signature)